

Unit 10 - Arcs and Angle Relationships in Circles

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.

- *Clarification 1: Within the Geometry course, problems are limited to relationships between inscribed angles; central angles; and angles formed by the following intersections: a tangent and a secant through the center, two tangents, and a chord and its perpendicular bisector.*

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.

- *Clarification 1: Instruction focuses on the conceptual understanding that the length of the arc intercepted by an angle is proportional to the radius.*

MA.912.GR.5.5 (Honors only) Given a point outside a circle, construct a line tangent to the circle that passes through the given point.

- *Clarification 1: When given a circle, the center must be provided.*
- *Clarification 2: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.*

Unit 10, Lesson 1: Sectors



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Targets

- ☐ I can define a sector of a circle.
- ☐ I can explain how the area of a sector is defined as a fractional part of the area of a circle.



10.1.1 Warm-Up: Pac-Man

The classic and popular Pac-Man video game was released in Japan in May of 1980 and was in the United States by October of that same year. The yellow, pie-shaped character travels around a maze trying to eat dots while avoiding the ghosts hunting him.

Watch the video of Pac-Man going around the maze and eating dots.



1. What do you notice about Pac-Man's body as he opens and closes his mouth to eat the dots?
2. What do you wonder as you watch Pac-Man move around the maze?



10.1.2 Exploration: Fractions of a Circle

Work with your partner to complete the following.

1. Pizza sizes are given in terms of the pizza's diameter. The standard size for most medium circular pizzas is 14 inches (in.) and it is usually cut into 10 triangular slices.



- a. What is the approximate area of a 14-inch pizza? Use $\frac{22}{7}$ for π .
- b. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 5 slices?

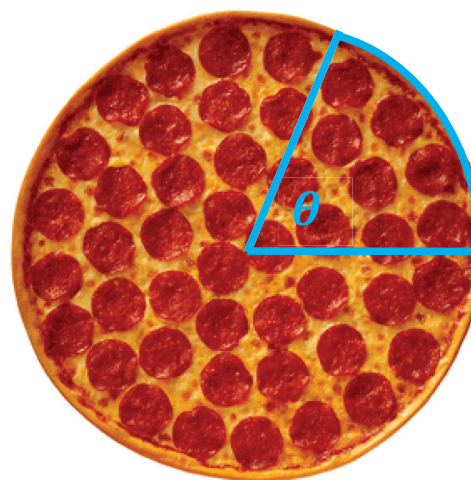
Fraction	Percentage

- c. What is the area of 5 slices?
- d. What is the area of 1 slice?

2. The point where a pizza is divided to create precisely equal slices is a central angle.



A **central angle** is an angle with its vertex at the center of a circle with radii as its side.



The central angle is a fraction of the total 360° in a circle.

- a. Complete the table. The first row of the table has been filled in.

	Fraction	Central Angle in Degrees ($^\circ$)
Half of a Circle	$\frac{1}{2}$	$\left(\frac{1}{2}\right) \cdot (360^\circ) = 180^\circ$
Quarter of a Circle		
Third of a Circle		
Tenth of a Circle		

b. Complete the statement.

One slice, or fraction of the circle, is bounded by two
☐ diameters
☐ radii and an arc. Since each slice is the same
 size, the central angle for each fractional piece of the
 circular pizza will ☐ vary.
☐ be equal.

c. The pizza slices are an example of **sectors**. With your partner, write a definition of the term **sector**.

d. Ask two other partner pairs to share their definition of a sector. Record a summary of their definition. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was correct.

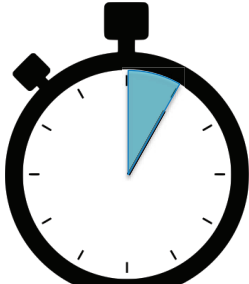
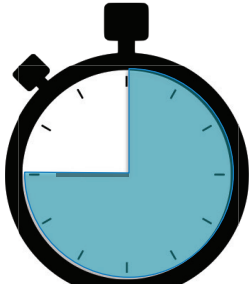
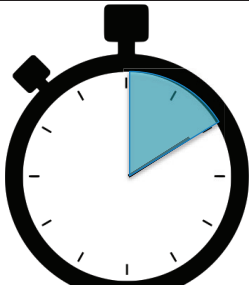
My Summary of Their Definition of a Sector	Initials



10.1.3 Guided Practice: Finding the Area of a Sector

The images below show a stopwatch with different time increments marked. The blue shading on the stopwatches each forms a sector.

1. Complete the table.

	Sector Fraction of the Circle	Central Angle Measure of the Sector	Ratio of the Central Angle to the Whole Circle
			
			
			

- Summarize what you notice about the relationship between the values in each row.

- Discuss with your partner how ratios could be used to find the area of the sectors, including any additional information that might be needed. Summarize your discussion.

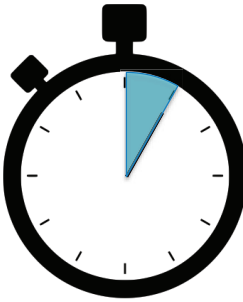
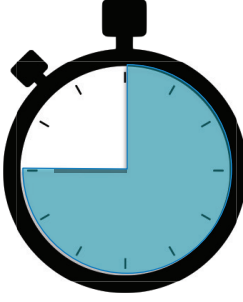
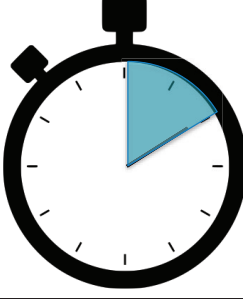
Given the radius of the stopwatch is 2 in., the area of the stopwatch is 4π in.².

$$A = \pi r^2$$

$$A = \pi(2)^2$$

$$A = 4\pi \text{ in.}^2$$

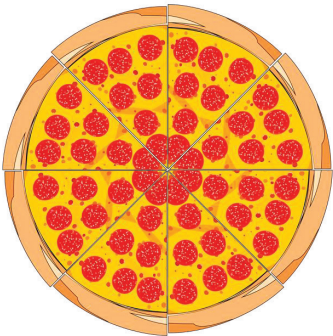
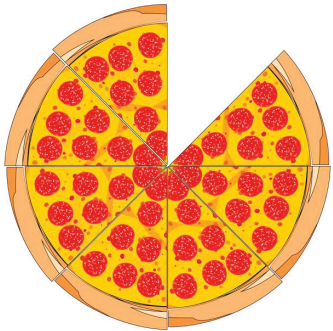
- Use the relationship between each sector and the whole stopwatch explored in problem 1 to find the area of each sector. Show your work.

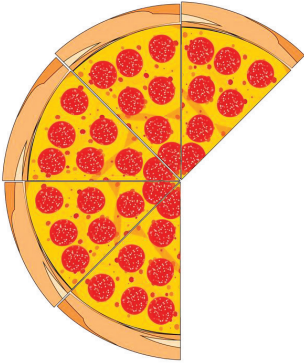
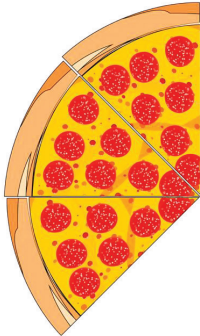
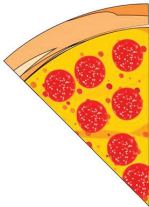
	Area of the Blue-Shaded Sector
	
	
	



10.1.4 Your Turn!

1. Kenya orders a large pizza that has a 16 in. diameter. Complete the table on the number of slices she has.

	Fraction of the Circle	Area of the Sector	Central Angle
Eight Slices 			
Seven Slices 			

	Fraction of the Circle	Area of the Sector	Central Angle
Five Slices 			
Three Slices 			
One Slice 			



10.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about sectors.

Unit 10, Lesson 2: Arc Length



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Targets

- ☐ I can define arc length.
- ☐ I can explain how the length of an arc is proportional to the ratio of the central angle to the whole circle, measured in degrees.



10.2.1 Warm-Up: Notice and Wonder

The Chandelier Tree, located in Leggett California, is part of a drive-thru tree park. The tree is believed to have been carved in the early 1930's by Charlie Underwood.



1. What do you notice?

2. What do you wonder?

3. Estimate the diameter of the tree at the base.

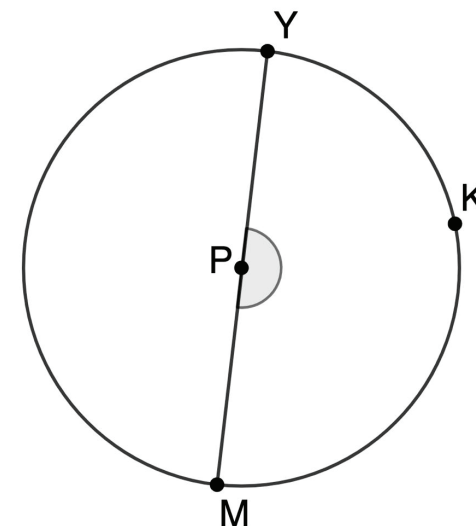
4. Explain why you chose that number as your estimate.



10.2.2 Exploration: Arc Length and Proportional Relationships

1. Circle P with point K on the circle and diameter $\overline{YM}$ is shown.

a. What is $m\angle YPM$?



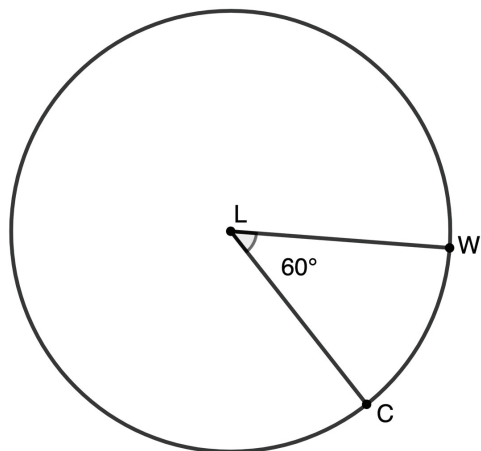
b. Determine the ratio of the measure of the central angle, $\angle YPM$, to the whole circle. Simplify the ratio.

c. An arc of a circle is the portion of the circumference between points. Point K is a point on Circle P . Discuss with your partner which part of the circle is arc YKM . Then, trace over this portion of the circle.

d. What fraction of the circle's circumference does arc YKM represent?

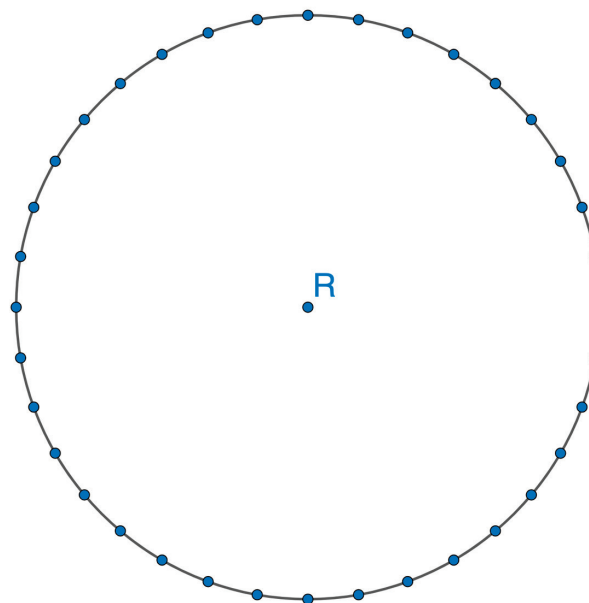
Arc YKM can also be notated as $\widehat{YKM}$.

2. Circle L is shown where points W and C are on the circle.



- What is the ratio of the central angle, $\angle CLW$, to the whole circle? Simplify the ratio.
 - What fraction of the circle's circumference does $\widehat{WC}$ represent?
3. Verify your answers to problems 1 and 2 with your partner and resolve any differences, if necessary.
4. With your partner, discuss any relationships you observe. Summarize your discussion.

5. On Circle R , the distance between each point measured along the circumference is 1 inch.



- Mark a point anywhere along the circle's circumference as point T . Then, starting at point T , mark an 8-inch section along the circle's circumference. Label the other endpoint of the 8-inch section as point G .
 - What fraction of the circle's circumference does the length of $\widehat{TG}$ represent?
- c. Draw the central angle $\angle TRG$.
- d. What is the measure of the central angle $\angle TRG$?

e. What is the ratio of the central angle to the whole circle?

f. Discuss with your partner the relationship between the length of the arc and the ratio of the central angle to the whole circle.

6. Complete the statements.

The

- arc length
- diameter
- radius
 is the distance along the part of a circle's circumference or along any curve. The ratio of the length of an arc to the circumference of a circle is

- the inverse of
- proportional to
 the ratio of the central angle to the whole circle.



10.2.3 Activity: Matching Central Angles and Ratios of Arc Length to Circumference

1. Your teacher will give you two sets of cards for this activity. One set shows circles with a central angle. The other set of cards is a set of ratios of the arc length to the circumference.
 - a. Work with your partner to match each circle card to the corresponding ratio indicated on the lettered card. Record the matches in the table.

Card Letter	Ratio of the Arc Length to the Circumference	Card Letter	Ratio of the Arc Length to the Circumference
A		F	
B		G	
C		H	
D		I	
E		J	

- b. Summarize how you and your partner were able to match the circles to a corresponding ratio without being given the circumference.

c. Complete the statement.

The ratio of the central angle to the whole circle is the

- ☐ inverse of
- ☐ same as

the ratio of the length of an arc to

the circumference.



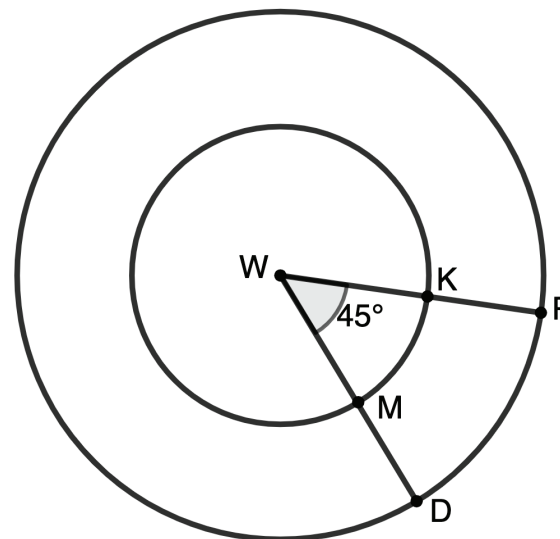
10.2.4 Exploration: Arc Lengths with the Same Central Angle and Different Radii

1. Points D and P lie on Circle W . The radius $\overline{WP}$ measures 5 inches (in.).

A smaller Circle W with a radius measure of 2 in. rests inside the larger Circle W , where points M and K lie on the smaller circle.

The smaller Circle W has central angle $\angle KWM$ that measures 45° and the larger Circle W has central angle $\angle PWD$ that measures 45° .

Use the information to complete the following.



- a. $\widehat{MK}$ and $\widehat{DP}$ are different lengths. Discuss with your partner whether you think the ratio of the arc length and circumference will be the same as or different than the ratio of the central angle of the whole circle.

- b. Andrew and Sarah are discussing if the ratio of the arc length to the circumference is the same as the ratio of the central angle to the whole circle for both the smaller and larger circles.

Sarah says, "Yes. They will have the same ratios because they have the same central angle."	Andrew says, "No. They do not have the same ratios because they are different circles with different radii."
---	--

Explain who you agree with.

- c. Complete the table to support your reasoning in Part B.

	Ratio of Arc Length to Circumference	Ratio of Central Angle to Whole Circle
Circle W with radius 2 in.		
Circle W with radius 5 in.		

- d. Explain whether your findings in Part C support your answers to Parts A and B. If not, explain why your answers to Parts A and B are incorrect.



10.2.5 Wrap-Up: Cool Down

- Write a summary of what you learned in this lesson.

Unit 10 Lesson 3: Radian Measure



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Targets

- ☐ I can define radian.
- ☐ I can explain the relationship between the length of the arc intercepted by an angle to the radius.



10.3.1 Warm-Up: Would You Rather?

Would you Rather...

Deposit \$5 in the bank and have it triple each week for 4 weeks?

OR

Deposit \$3 in the bank and have it quadruple each week for 4 weeks?

1. Which would you rather choose? Justify your reasoning using mathematics.



10.3.2 Exploration: Discovering Radian Measure

Angles that are measured in degrees can also be measured in **radians**. Measuring in radians is similar to measuring distance using other units, such as inches or feet. Radians can be written as a decimal or in pi (π) form.

With your partner, work through the Desmos activity "Discover Radian Measure." Complete the steps to answer each question.

Take a moment to look over the activity and the information provided.

- Slider " r " adjusts the radius of the circle. The radii for the activity range from 0 to 3, increasing by 0.5.
- Slider " c " adjusts the central angle.
- The central angle, $C_{centralAngle}$, is measured in radians, written as a decimal. Radians for the activity increase by 0.25.

Drag slider r to the right and left to observe how the measurements of radius, central angle, and arc length change as the r value changes. These values are in rows 2, 5, and 6 respectively of the activity.

Hit play on slider c to watch how the measurements of radius, central angle, and arc length changes. These values are in rows 2, 5, and 6 respectively.

1. Adjust slider c , $C_{centralAngle}$, so that the arc length, $A_{rclength}$, equals the radius, r . Complete the table for three different measures of r .

Radius (r)			
$C_{centralAngle}(c)$			
$A_{rclength}$			

2. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

3. Adjust slider r , radius, so that the arc length, $A_{rclength}$, is twice the radius, r . Complete the table for three different measures of r .

Radius (r)			
$C_{centralAngle}(c)$			
$A_{rclength}$			

4. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

5. Complete the statements based on your work in questions 1-4.

A radian is a unit used to express the measure of an

angle. Radians are based on the number of ☐ angles
☐ radii

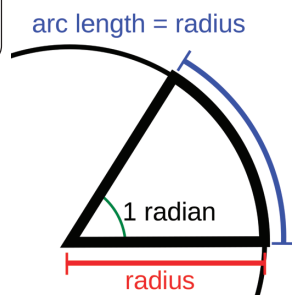
of a circle being wrapped along the ☐ diameter
☐ circumference

of the circle. The radian measure of a central angle is

equal to the length of the ☐ arc
☐ radius

intercepted by the central angle divided by the

☐ radius.
☐ arc length.

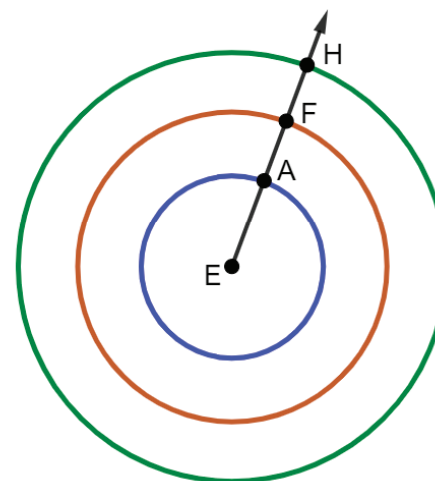


6. Describe how the diagram shows the relationship between the radius and arc length.



10.3.3 Exploration: Discovering Radians in a Whole Circle

1. Consider the diagram depicting three concentric circles each with center E . Point A lies on the blue Circle E , and has radius $EA = 1$ unit. Point F lies on the brown Circle E , and has radius $EF = 2$ units. Point H lies on the green Circle E , and has radius $EH = 3$ units.



- a. Complete the table with your partner.

Circle	Circumference	Ratio of Circumference to Radius
Radius $EA = 1$ unit		
Radius $EF = 2$ units		
Radius $EH = 3$ units		

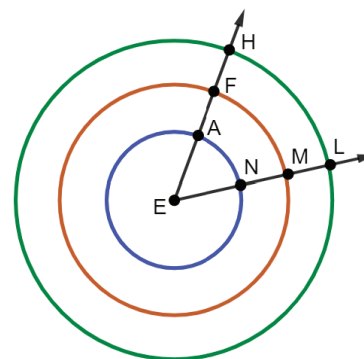
- b. Discuss any observations you made about the values in the table with your partner. Summarize your discussion.

- c. Write a conjecture about the ratio of the circumference to radius and a whole circle measured in radians.



10.3.4 Bring It Together: Relating Arc Length to the Radius

1. Concentric circles with center E are shown. Points A and N lay on the blue Circle E , and has radius $EA = 1$ unit. Points F and M lay on the brown Circle E , and has radius $EF = 2$ units. Points H and L lay on the green Circle E , and has radius $EH = 3$ units.



Describe how the arclengths AN , FM and HL are related.

2. Use words from the bank below to complete the statements.

angle arc radian radius

The length of the _____ intercepted by a(n) _____ is proportional to the _____ of the circle. The _____ measure of the angle is the constant of proportionality.



$$\text{Radians} = \frac{\text{Circumference}}{\text{Radius}} = \frac{2\pi r}{r} = \frac{2\pi}{1}$$

$$360^\circ = 2\pi \text{ radians}$$

$$180^\circ = \pi \text{ radians}$$



10.3.5 Wrap-Up: Cool Down

1. Your friend was absent today and will need notes from class. Write a quick note to help them understand relationship between the length of an arc intercepted by an angle and the radius.

Unit 10, Lesson 4: Solving Problems Involving Arc Length and Area of a Sector



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Target

- ☐ I can solve mathematical and real-world problems involving arc length and sector area.



10.4.1 Warm-Up: Bell Work

1. A small, family-owned restaurant sells a personal pan pizza with a radius of 10 centimeters for \$5. A nearby restaurant cuts their pizza with a radius of 30 centimeters into 8 slices of equal size, and charges \$5 per slice.

Which is a better deal? Explain your reasoning using mathematics.



10.4.2 Exploration: Finding a Formula for the Area of a Sector

1. Suppose a circle is cut into 8 equal slices to form sectors.

a. Complete the table.

Number of Slices	Fraction of the Circle	Central Angle in Degrees ($^{\circ}$)	Write an Equation to Find the Area of the Sector
2			
4			
6			
8			

- b. Compare the values in your table with your partner. Discuss your equations in the last column and make any revisions if needed.

- c. Discuss with your partner how the fractions in your equations relate to the central angles, and how the equations might have differed if the angle measure was given in radians.

- d. Summarize your discussion by writing a general formula to find the area of any sector.



Area of a Sector

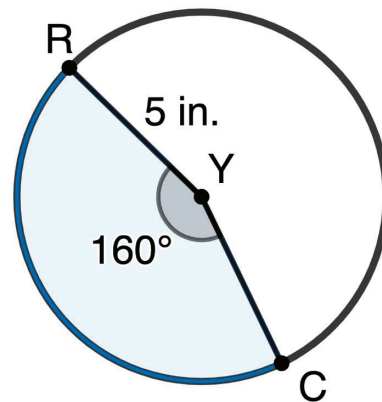
$$A = \frac{\theta}{360} \pi r^2, \text{ where } \theta \text{ is in degrees}$$

$$A = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$



10.4.3 Guided Practice: Area of a Sector

1. Circle Y has radius $\overline{CY}$ that measures 5 inches with central angle $\angle RYC$, where $m\angle RYC = 160^\circ$. Find the area of the shaded sector. Round your answer to the nearest thousandth.



2. Circle X has a diameter of 12 centimeters. Points B and P are on the circle and form the sector BXP . The central angle $\angle BXP$ in Circle X measures 60° . Determine the area of the sector. Express your answer in terms of π .
3. The area of a sector with a radius of 8 inches is 74.84 square inches. Calculate the approximate measure of the central angle of the sector in degrees.



10.4.4 Collaboration: Arc Length

Work with your partner to complete the following.

1. Explain how the circumference relates to an arc length of a circle.
2. Describe how to find the arc length of half a circle.
3. Describe how to find the arc length of a quarter of a circle.
4. Describe how to find the arc length of a circle with a central angle of 60° .

5. Write a general equation that can be used to find the arc length of a sector in degrees.

6. Discuss with your partner how your equation would change if the central angle had been given in radians. Summarize your discussion by writing an equation to find the arc length if the angle measure is given in radians.



Arc Length

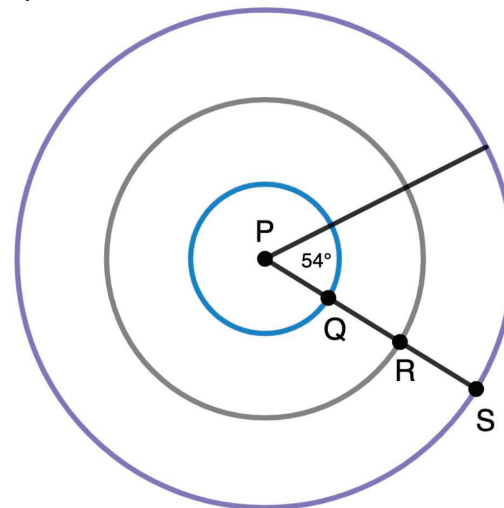
Arc Length = $C \cdot \frac{\theta}{360^\circ}$, where θ is in degrees

Arc Length = $r \cdot \theta$, where θ is in radians



10.4.5 Guided Practice: Arc Length

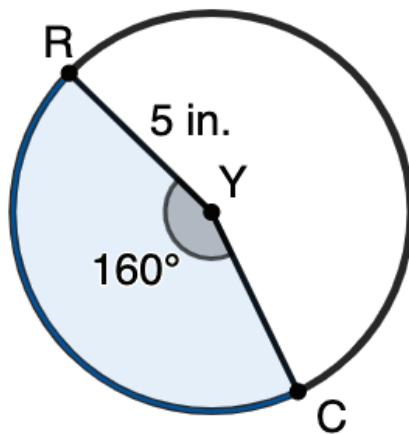
1. Concentric circles with center P are shown with $m\angle P = 54^\circ$. The smallest Circle P has a radius $\overline{PQ}$ that measures 15 meters (m). The largest Circle P has radius $\overline{PS}$ and the middle Circle P has radius $\overline{PR}$. Each circle has a radius that is 15 m longer than the immediate prior circle.



Calculate the arc length intercepted by $\angle P$ on each circle.

Arc length of circle with radius $\overline{PQ}$	
Arc length of circle with radius $\overline{PR}$	
Arc length of circle with radius $\overline{PS}$	

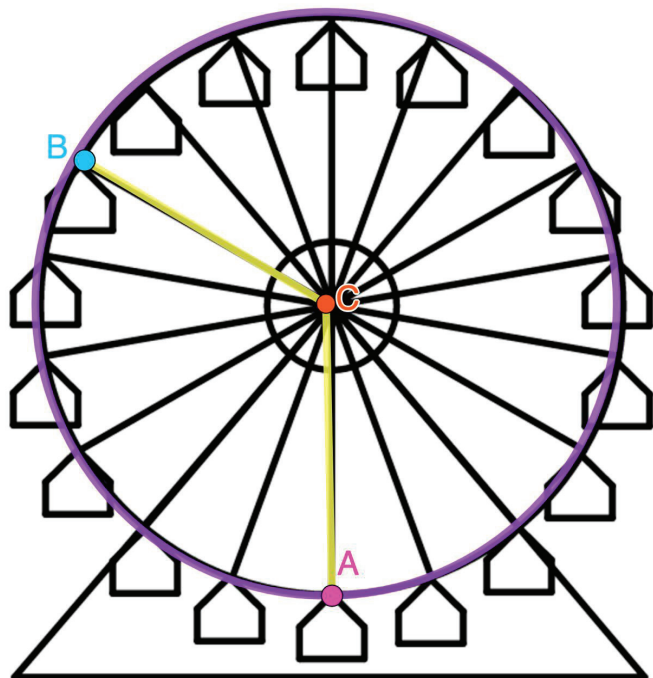
2. Points R and C are on Circle Y . If $m\angle CYR$ is 160° , find the length of $\widehat{RC}$.



10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector

1. In Sarah's family, the person celebrating their birthday always gets to cut the first piece of a circular birthday cake. Sarah is celebrating her birthday with both of her parents, her two brothers and her best friend. She cuts her piece of birthday cake and it formed a sector with a diameter of 8 inches and a central angle with measure 85° . If the 5 other party guests equally share the remaining portion of the cake, what is the approximate area of cake each person receives? Round your answer to two decimal places.

2. Dominic is riding a Ferris wheel at the county fair. The Ferris wheel forms a circle with center C . The Ferris wheel has a radius of 60 ft and rotates clockwise with seats that are equidistant from each other.



- a. After the ride begins from point A and travels **clockwise**, it pauses at point B . How many feet has Dominic traveled when his seat pauses at point B ? Write your answer in π form and rounded to the nearest thousandth.

π or feet

- b. About how many feet has Dominic traveled if the full ride is 4 revolutions?

π or feet



10.4.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 10, Lesson 5: Solving Problems Involving Arc Measure and Central Angles



Benchmark

MA912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



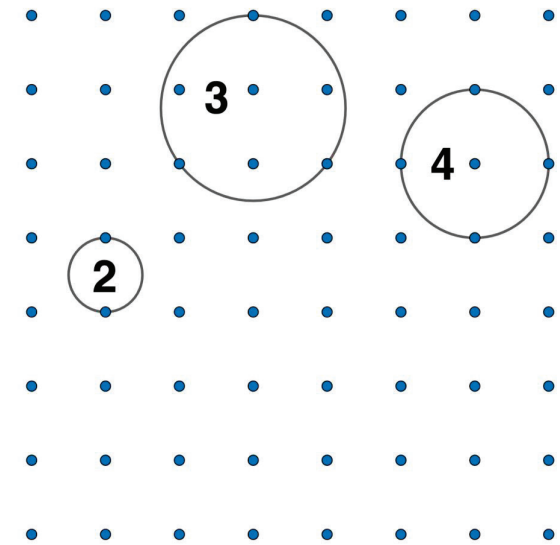
Learning Target

- ☐ I can use the relationship between the measure of a central angle and its intercepted arc to solve problems.



10.5.1 Warm-Up: Circles

1. A square lattice with three circles is shown, where the circles pass through exactly 2, 3, or 4 points on the lattice, respectively.

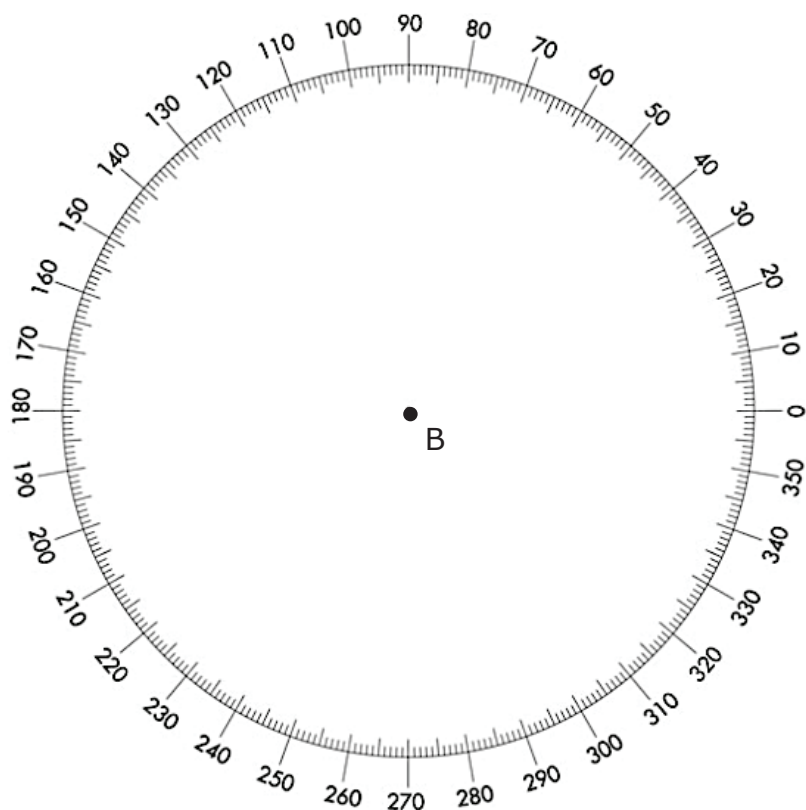


- a. What do you notice about the circles?
- b. Connect the points on each circle to create a polygon. Describe any pattern you notice.
- c. What type of polygon would be inside a circle that passes through 5 points?
- d. On the lattice, draw a circle that passes through exactly 5 points.



10.5.2 Exploration: Understanding Central Angles

1. Circle B is shown.



- Find the 130° mark and label it point Y . Then, find the 220° mark and label it point L . Draw along the circumference from point Y to point L going counterclockwise.
- Exactly what fraction of the circle's circumference is from point Y (130°) to point L (220°)?

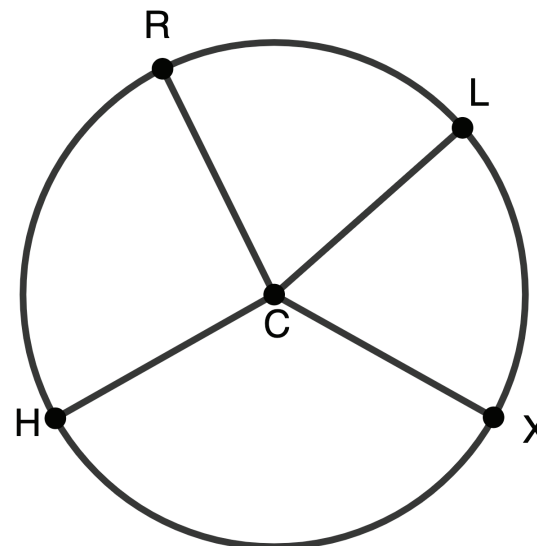
- Draw the radius $\overline{BY}$ from the center B to point Y . Then, draw another radius $\overline{BL}$ from the center B to point L .
- What is the measure, in degrees, of the central angle $\angle YBL$ in relation to $\widehat{YL}$?
- Next, along the circumference, label 0° as point U , 90° as point S , 250° as point E , and 340° as point H .
- Draw two separate arcs along the circumference of the circle, one extending from point U to point S and the other extending from point E to point H . Then, draw the central angles that intersect with each part of the circumference you marked.
- What is the measure, in degrees, of each of the central angles, $\angle UBS$ and $\angle EBH$?
- Describe what you notice about all three central angles.
- Explain the relationship between a central angle and its intercepted arc.

- j. Share your response with your partner. Listen to your partner's explanation and write a summary of the explanation. Have your partner initial stating that your summary of their explanation is accurate.

Your Partner's Explanation	Partner's Initials

- k. Draw a diagram that shows the relationship between a central angle and its intercepted arc.
- l. Discuss with your partner how arc measure relates to the relationship between a central angle and its intercepted arc. Summarize your discussion.

2. Circle C is shown where points R , L , X , and H are on the circle.



- a. Name the angle that intercepts $\widehat{RH}$.
- b. Name the arc that is intercepted by $\angle HCX$.

- c. Brooklynn and Angelina were discussing how to name an arc that is intercepted by $\angle RCX$. Each of their explanations are shown.

Brooklynn	Angelina
The arc that is intercepted by $\angle RCX$ is $\widehat{RX}$ because you just use the first and last letters of the angle to name the arc.	An arc that is intercepted by $\angle RCX$ could be $\widehat{RLX}$ or $\widehat{RHX}$ because the angle intercepts two arcs.

Discuss with your partner the conversation between Brooklynn and Angelina. Summarize your discussion.



10.5.3 Bring It Together: Arc Measure and Central Angles

An **arc** of a circle is a section of the circumference of the circle.



The **length of an arc** is simply the length of the section of the circumference.



In a circle, the **measure of an arc** (in degrees or radians) is equal to the measure of the central angle that intercepts the arc.

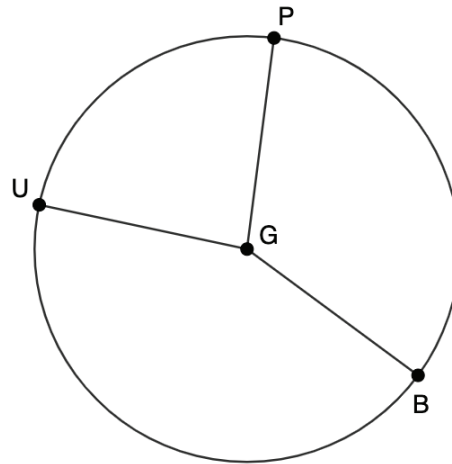


In a circle, any arc whose corresponding central angle has a measure greater than 180° is considered a **major arc** and should be named using three letters.



In a circle, any arc whose corresponding central angle has a measure less than 180° is considered a **minor arc** and should be named using two letters.

1. Circle G has points P , B , and U on the circle with $m\angle PGB = 120^\circ$ and $m\widehat{BU} = 155^\circ$. Ryan and Katelyn are finding the measure of $\widehat{UP}$, but they found different answers.



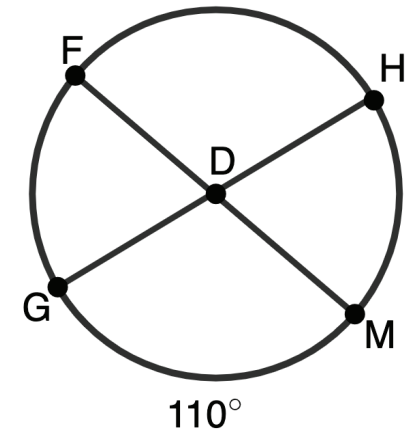
Ryan	Katelyn
$155 + 120 = 275$	$155 + 120 = 275$
$275 - 180 = 95$	$360 - 275 = 85$
$m\widehat{UP} = 95^\circ$	$m\widehat{UP} = 85^\circ$

Do you agree with Ryan or Katelyn? Justify your answer.



10.5.4 Guided Practice: Finding Angle Measures

1. Circle D has diameters $\overline{FM}$ and $\overline{GH}$ that both intersect at Point D . Given the $m\widehat{MG} = 110^\circ$, find the $m\widehat{HF}$.



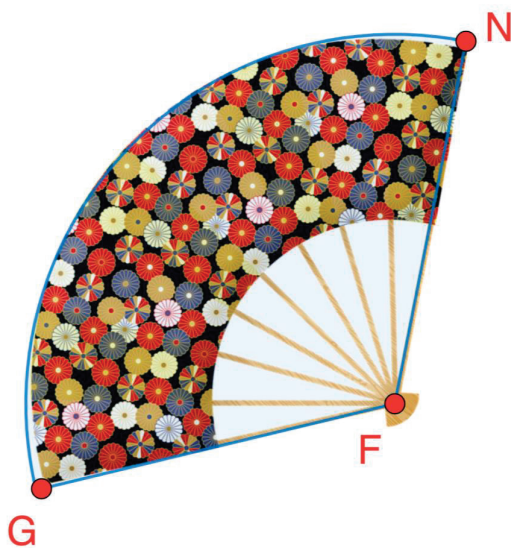
2. Circle T has points A , V , and C on the circle with $m\angle VTA = 21.7^\circ$. If $m\widehat{VC} = 132.3^\circ$ and $\widehat{VC}$ does not intersect $\widehat{VA}$, what is $m\angle CTA$?

3. Circle C has radii $\overline{CN}$ and $\overline{CL}$. Find the value of x if $m\angle LCN = 114^\circ$ and $m\widehat{LN} = (6x + 6)^\circ$.

4. Isabella is creating a folding fan for art class. She needs to determine the measure of the arc of the fan before she can cut the paper.

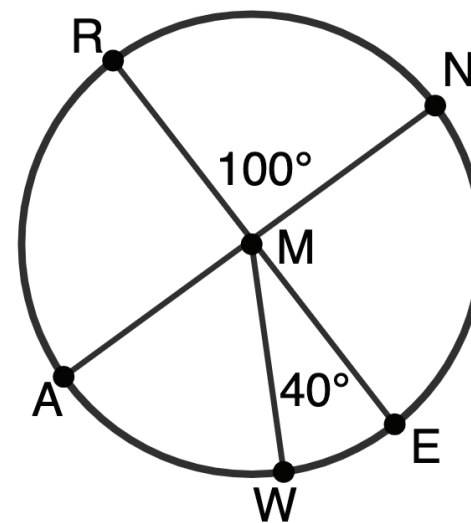
She uses the expression $(8x + 102)^\circ$ as the measure of the central angle, $\angle NFG$. For the $m\widehat{GN}$, she uses the expression $(32x + 54)^\circ$.

What is the degree measure of the arc of the fan, $\widehat{GN}$?



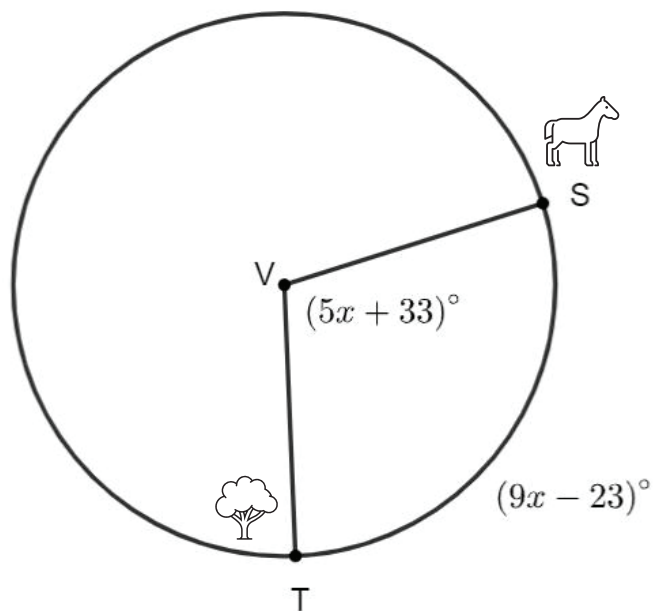
10.5.5 Your Turn!

1. In Circle M , $m\angle RMN = 100^\circ$ and $m\angle EMW = 40^\circ$. Calculate $m\widehat{RNW}$ given that M is the center and $\overline{RE}$ and $\overline{AN}$ are diameters.



2. Spirit is tied to a post at point V and is grazing in a circular pattern. He decides to walk from his position at point S to the tree at point T along the circumference of the circular area he can roam.

If the $m\angle SVT = (5x + 33)^\circ$ and $m\widehat{ST} = (9x - 23)^\circ$, find the measure of the arc that Spirit walked, $m\widehat{ST}$.



10.5.6 Wrap-Up: Ticket Out the Door

1. Circle X has points R , P , and A on the circle. The measure of the central angle $\angle AXP = 50^\circ$, $m\widehat{RA} = 141^\circ$, and $\widehat{RA}$ does not intersect $\widehat{AP}$. Find $m\angle RXP$.

Unit 10, Lesson 6: Solving Problems Involving Inscribed Angles



Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



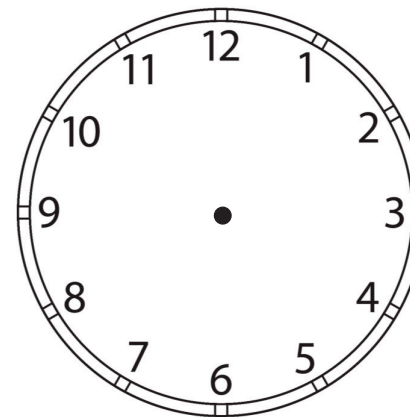
Learning Target

- ☐ I can use the relationship between the measure of an inscribed angle and its intercepted arc to solve problems.



10.6.1 Warm-Up: Angles on a Clock

1. Use the clock to answer the following questions.



- a. If the time is 10:10 and the hour hand does not move, what is the exact angle between the minute hand and hour hand?
- b. If the hour hand does move, what is the exact angle if the time is 7:30?
- c. If the hour hand does move, what is the exact angle if the time is 7:35?



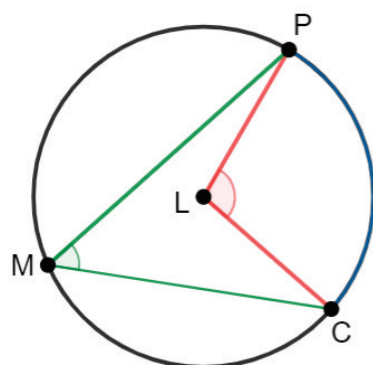
10.6.2 Exploration: Inscribed Angles

- Circle L is shown where P , C , and M are points on the circle.

A **chord** is a line segment on the interior of a circle with both endpoints lying on the circle.



An **inscribed angle in a circle** is an angle which is formed in the interior of a circle when two chords share an endpoint.



Central Angle $m\angle PLC = 110^\circ$

Inscribed Angle $m\angle PMC = 55^\circ$

Arc Measure $\widehat{PC} = 110^\circ$

- What is the connection between the central angle and the arc measure?
- Describe the relationship between the central angle and the inscribed angle.

- Write a conclusion about the measures of the central angle, the inscribed angle, and the arc measure.

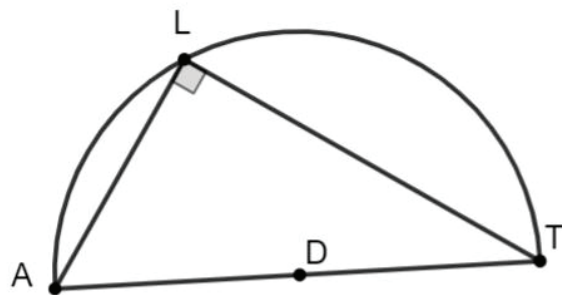
- Share your response to Part C with two classmates. Listen to their conclusion and summarize it in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating your summary was correct.

My Summary of Their Conclusion	Initials

- Using Circle L , complete the table.

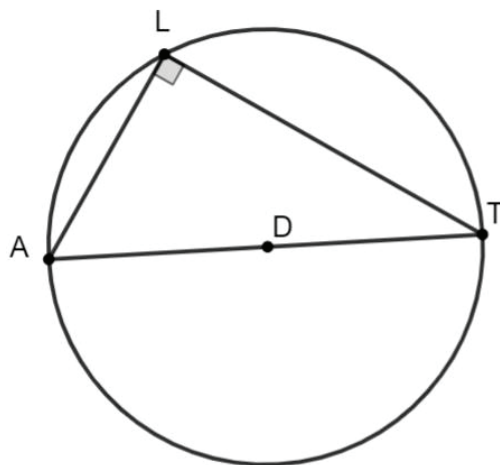
Central Angle	Inscribed Angle	Arc Measure
216°		
	81°	
		143°
	145°	

2. $\angle ALT$ is inscribed in semicircle D , as shown.



- a. Describe the angle inscribed in the semicircle.

- b. If the entire Circle D is shown, what is the measure of $\widehat{AT}$?



- c. Discuss with your partner the relationship between an inscribed angle and a semicircle.



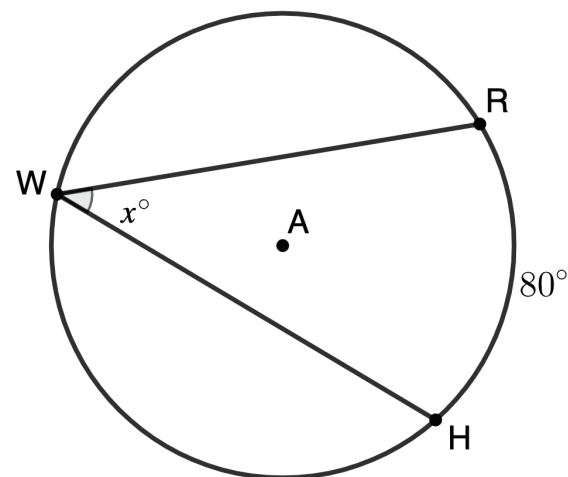
Inscribed Angle Theorem

The measure of an inscribed angle is one-half the measure of the intercepted arc.

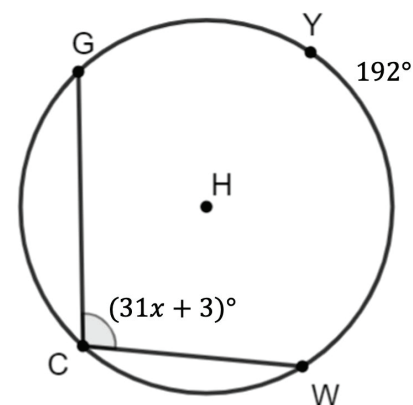


10.6.3 Guided Practice: Inscribed Angles

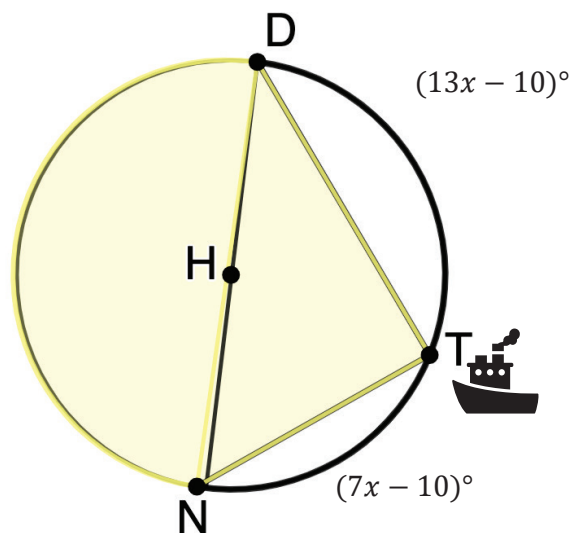
1. Circle A is shown where points R , W , and H are on the circle. $\angle RWH$ is inscribed inside Circle A with $m\widehat{RH} = 80^\circ$. Find the value of x .



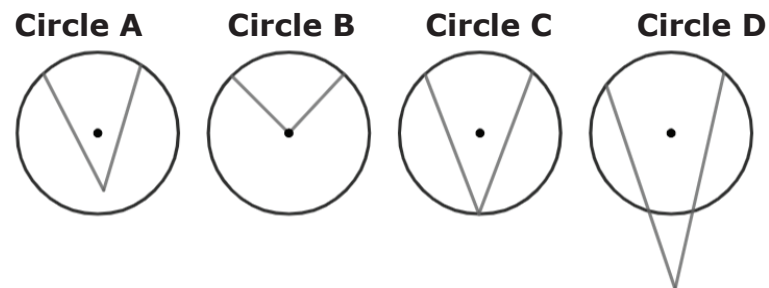
2. Points Y , G , C , and W lie on Circle H . $\angle GCW$ is inscribed in the circle. Find x , given $m\widehat{GYW} = 192^\circ$ and $m\angle GCW = (31x + 3)^\circ$.



3. Boats use angle measures and arcs when sailing in treacherous waters. When a boat captain turns on the boat light, it forms inscribed angle $\angle NTD$. Use the information for Circle H to find $m\angle NDT$.



4. Craig and Valentina are reviewing circles to identify inscribed angles.



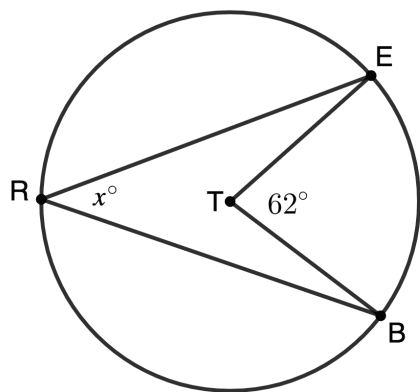
Valentina identified Circle C is an inscribed angle. Craig identified both Circle A and Circle C as inscribed angles.

Do you agree with Valentina, Craig, or neither? Justify your reasoning.

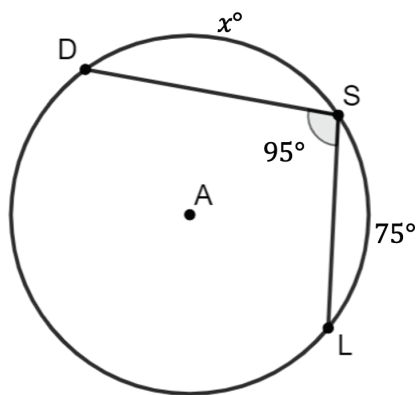


10.6.4 Your Turn!

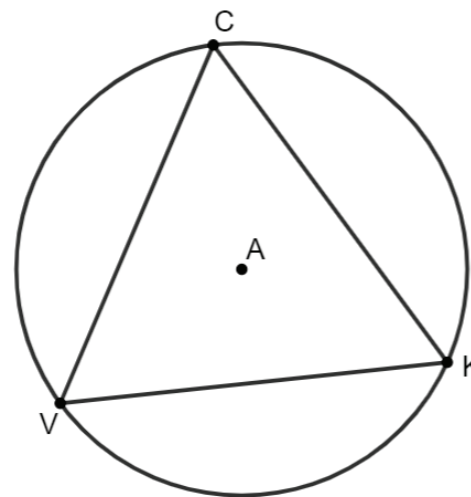
1. Circle T is shown where $\angle ETB$ is a central angle, $\angle ERB$ is inscribed in the circle, and $m\angle ETB = 62^\circ$. Find $m\angle ERB$.



2. Circle A has points D , S , and L on the circle. $\angle DSL$ is inscribed in Circle A measuring 95° . Given $m\widehat{SL} = 75^\circ$, find $m\widehat{DS}$.



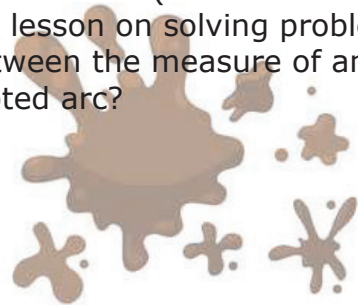
3. Equilateral triangle CKV is inscribed in Circle A . Circle A consists of three equal inscribed angles. Determine the value of the inscribed angle.





10.6.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on solving problems using the relationship between the measure of an inscribed angle and its intercepted arc?



Unit 10, Lesson 7: Solving Problems Involving Intersecting Chords



Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



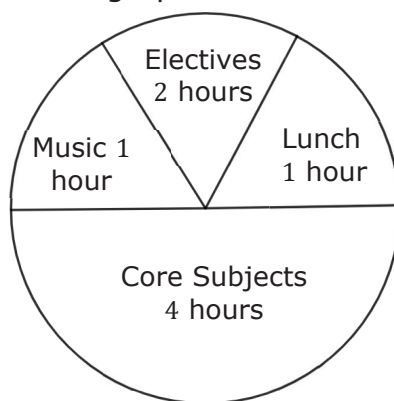
Learning Target

- ☐ I can solve problems involving chords of a circle and their related angles.



10.7.1 Warm-Up: Sectors

1. What do you notice about the circle graph?

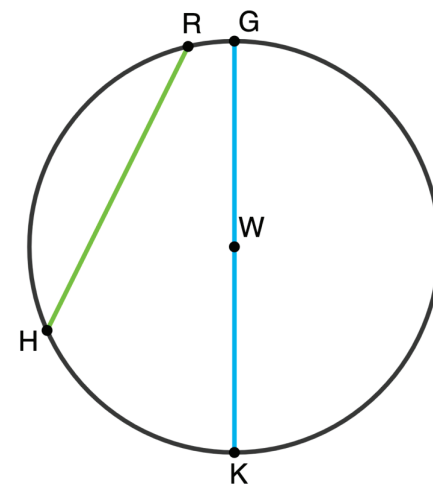


2. Identify a sector that is incorrectly sized.
3. Redraw the circle graph so that the sizes are proportional to the number of hours.



10.7.2 Exploration: Chords

Circle W is shown where points R , G , K , and H lie on the circle.



1. What similarities do you notice between $\overline{RH}$ and $\overline{GK}$?
2. What differences do you notice between $\overline{RH}$ and $\overline{GK}$?

A **chord** is a line segment that connects two points on the curve of a circle.

3. Complete the statements.

$\overline{RH}$ is

- ☐ a diameter
- ☐ a chord
- ☐ both a diameter and a chord

of the circle.

$\overline{GK}$ is

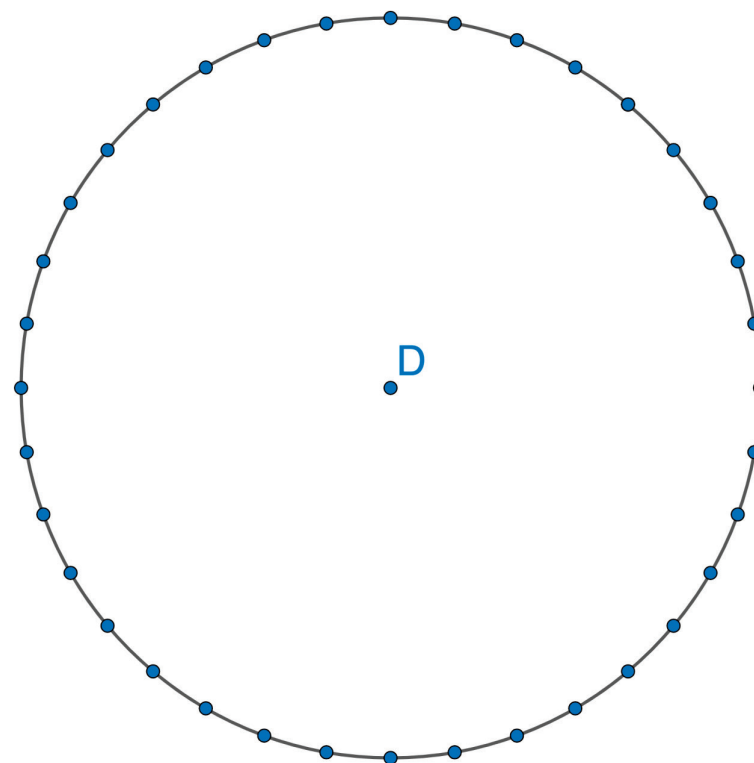
- ☐ a diameter
- ☐ a chord
- ☐ both a diameter and a chord

of the circle.



10.7.3 Collaboration: Angles in Intersecting Chords

1. With your partner, complete the following using Circle D.
 - a. Draw and label two chords that intersect inside the circle, making sure the endpoints fall on the points. When drawing the chords, do not draw the same chords as your partner and avoid drawing diameters.



- b. Choose one of the angles to measure and label. Then, complete the table.

	Angle Measure	Measure of Arc it Intercepts	Measure of the Arc the Vertical Angle Intercepts
Your Angle			
Your Partner's Angle			

- c. Discuss the following with your partner.
- What do you notice about the values in the table?
 - What do you think the relationship is between the angle measure and the two intercepted arcs?
- d. Write a formula that generalizes your discovery from the discussion you had in Part C.

- e. Complete the statement.



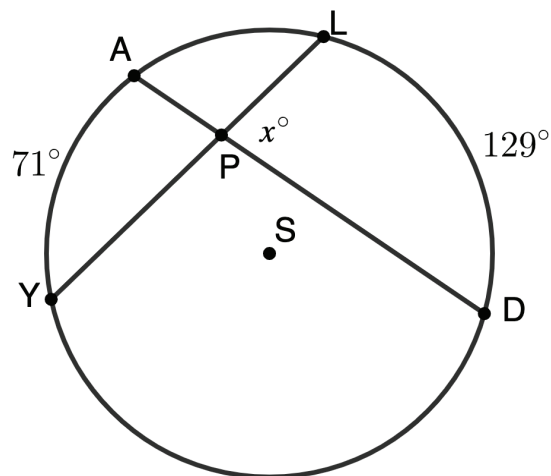
Angles of Intersecting Chords Theorem

The angle formed inside of a circle by two intersecting chords is ☐ double ☐ equal ☐ half the sum of the measures of the arcs intercepted by the angle and its vertical angle.

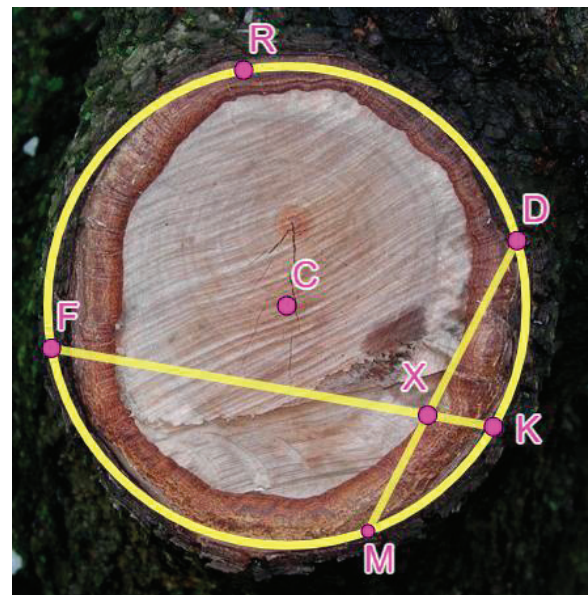


10.7.4 Guided Practice: Solving Problems Involving Angles in Intersecting Chords

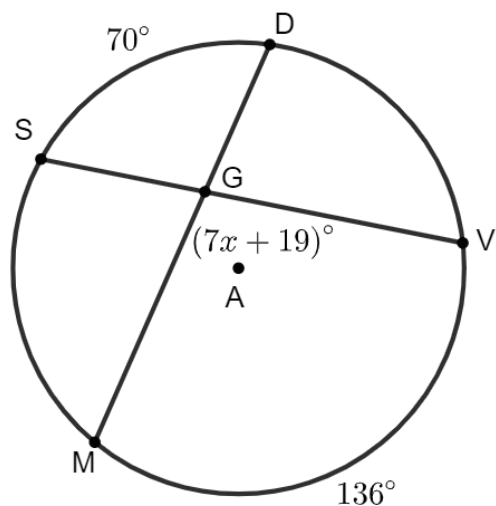
1. Circle S is shown, where chords $\overline{AD}$ and $\overline{YL}$ intersect at point P . Find the value of x for $\angle LPD$, given $m\widehat{AY} = 71^\circ$ and $m\widehat{LD} = 129^\circ$.



2. A tree is cut down and a circular tree stump is left. The top of the tree stump has two cracks that represent two chords, $\overline{FK}$ and $\overline{DM}$, intersecting at point X in Circle C . If $m\widehat{FRD} = 175^\circ$ and $m\widehat{K\widehat{M}} = 39^\circ$, what is $m\angle FXD$?

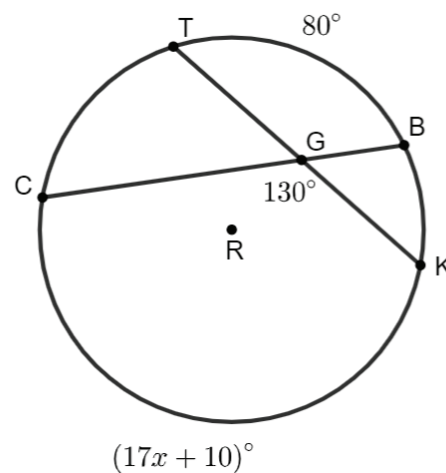


3. Circle A is shown, where chords $\overline{SV}$ and $\overline{MD}$ intersect at point G . Find the value of x , given $m\widehat{SD} = 70^\circ$, $m\widehat{MV} = 136^\circ$, and $m\angle MGV = (7x + 19)^\circ$.



10.7.5 Your Turn!

1. Circle R is shown, where chords $\overline{TK}$ and $\overline{CB}$ intersect at point G . Find the value of x , given $m\widehat{TB} = 80^\circ$, $m\widehat{CK} = (17x + 10)^\circ$, and $m\angle CGK = 130^\circ$.





10.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to solving problems involving chords.

1. I can define a chord of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can define the relationship of angles and arcs formed by the intersection of two chords.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. I can solve problems involving chords of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.



Unit 10, Lesson 8: Solving Problems Involving Intersection of Secants



Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



Learning Target

- ☐ I can solve problems involving secants drawn from a point external to the circle and their related angles.



10.8.1 Warm-Up: Satellite Operations Engineer

- Veronica Navarro is a satellite operations engineer. Before beginning her career, she received a bachelor's degree. One piece of advice she would give to high school students is to take as many classes in high school as possible because they are free. In her opinion, a degree beyond high school is needed in many careers because employers want to know a person can be trained and is capable of learning. Veronica's employer paid for her to receive additional degrees as a means to further develop her abilities and deepen her learning.

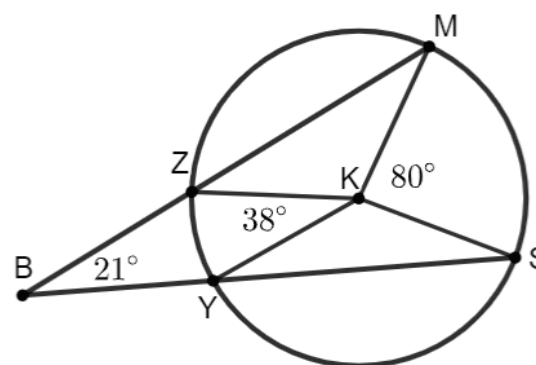
Why do you think it is important for companies to know you are capable of being trained?



10.8.2 Exploration: Angles Formed by Secants

A **secant** is a line or line segment that passes through two points on a circle. In the interior of the circle, a secant line segment forms a chord.

- $\angle MBS$ intersects Circle K , creating two intersecting arcs, where points M , Z , Y , and S are on the circle. The $m\angle ZKY = 38^\circ$, $m\angle SKM = 80^\circ$, and $m\angle MBS = 21^\circ$.



- Complete the table.

Arc	Central Angle Measure	Arc Measure
$\widehat{MS}$		
$\widehat{ZY}$		

- Add and subtract the two arc measures.

$$m\widehat{MS} + m\widehat{ZY} = \underline{\hspace{2cm}}$$

$$m\widehat{MS} - m\widehat{ZY} = \underline{\hspace{2cm}}$$

- c. Do the operations in Part B compare to the measure of $\angle MBS$? If so, explain.

- d. With your partner, write a formula that can be used to find the measure of an exterior angle of two secant lines.



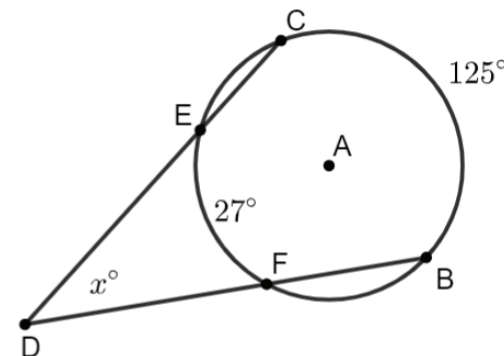
Exterior Secant Angle Theorem

The measure of an angle formed by two secants intersecting in the exterior of a circle is one-half the difference of the measures of the intercepted arcs.



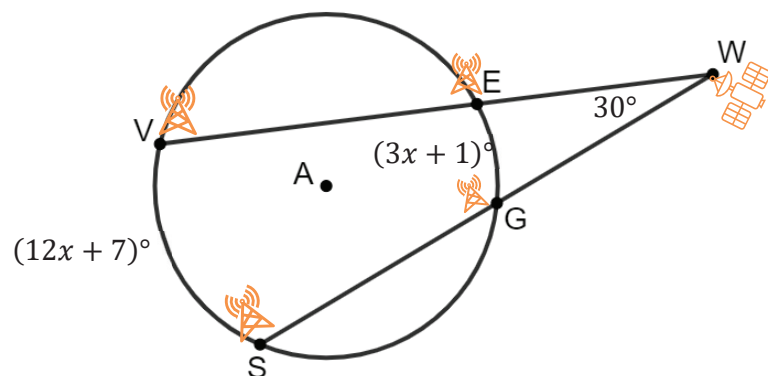
10.8.3 Guided Practice: Solving Problems Involving Angles Formed by Secants

1. Circle A has points C , E , F , and B on the circle. Secants $\overline{CD}$ and $\overline{BD}$ intersect at point D outside the circle. $m\widehat{CB} = 125^\circ$, $m\widehat{EF} = 27^\circ$, and $m\angle CDB = x^\circ$. What is the value of x ?



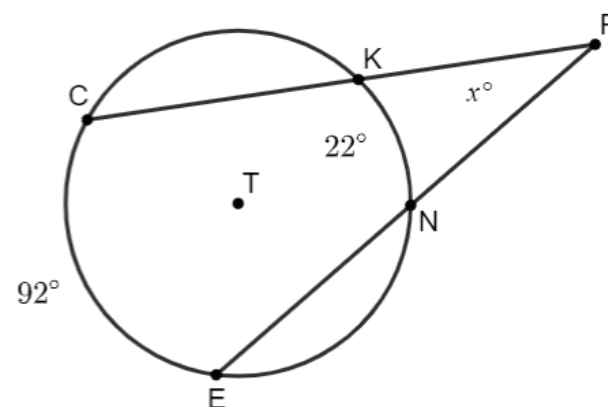
2. A satellite, point W , sends signals $\overline{VW}$ and $\overline{SW}$ to Earth. Signal $\overline{VW}$ hits tower E before reaching tower V . Similarly, signal $\overline{SW}$ hits tower G before reaching tower S . Scientists are interested in finding the measure of the intersecting arcs created by the two signals. They know $m\angle W = 30^\circ$, $m\widehat{VS} = (12x + 7)^\circ$, and $m\widehat{EG} = (3x + 1)^\circ$.

Find $m\widehat{VS}$ and $m\widehat{EG}$.



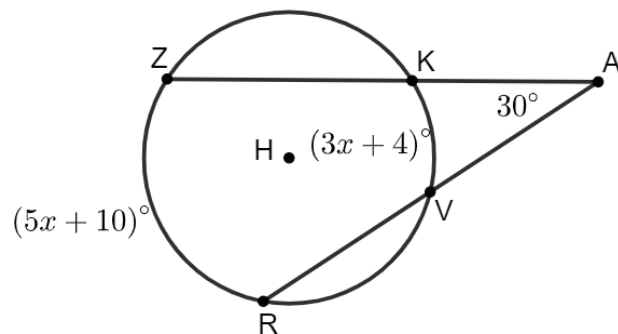
10.8.4 Your Turn!

1. Circle T has points C , K , N , and E on the circle. Secants $\overline{CR}$ and $\overline{ER}$ intersect at point R outside the circle. The $m\widehat{CE} = 92^\circ$, $m\widehat{KN} = 22^\circ$, and $m\angle CRE = x^\circ$. What is the value of x ?



2. Circle H has points Z , K , V , and R on the circle. Secants $\overline{ZA}$ and $\overline{RA}$ intersect at Point A outside the circle. The $m\widehat{ZR} = (5x + 10)^\circ$, $m\widehat{KV} = (3x + 4)^\circ$, and $m\angle ZAR = 30^\circ$.

Find the measure of $\widehat{ZR}$.



10.8.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson.

Unit 10 Lesson 9: Solving Problems Involving Tangents



Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



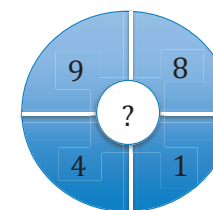
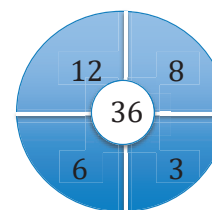
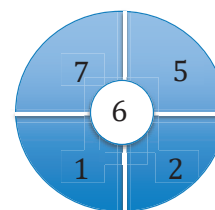
Learning Target

- ☐ I can solve problems related to tangents and secants and the measures of their intercepted arcs.



10.9.1 Warm-Up: Number Puzzle

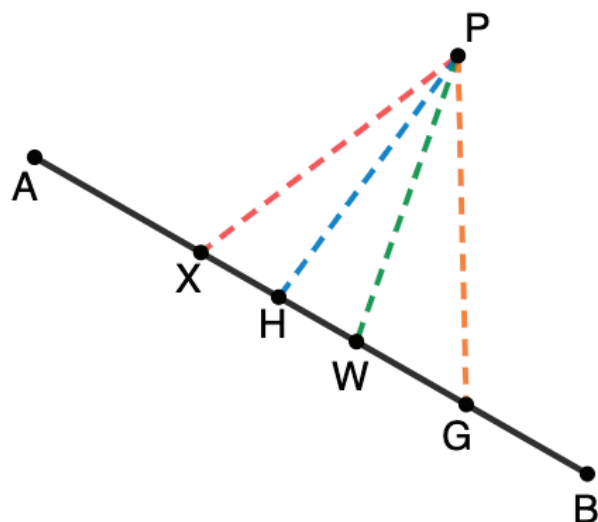
- In the circles, the outer numbers are used to calculate the number in the middle. Find the rule to determine what number should replace the question mark in the third circle.





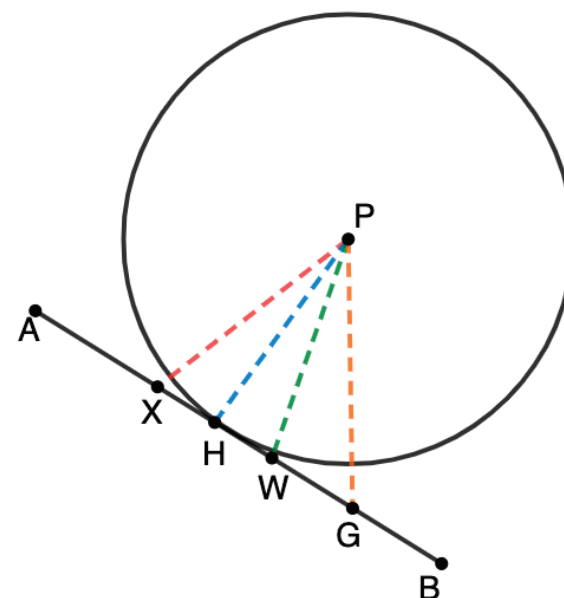
10.9.2 Exploration: Intersecting Tangent Lines

1. With your partner, answer the following.



- a. Without measuring, which dashed line represents the shortest distance from point P to $\overline{AB}$? Explain your answer.

- b. Point P is the center of a circle that intersects $\overline{AB}$. In the table, describe your observations of each dashed segment in relation to the center of Circle P .



Segment	Relationship with the Center of Circle P
$\overline{PG}$	
$\overline{PW}$	
$\overline{PH}$	
$\overline{PX}$	

- c. Discuss with your partner the relationship the blue segment has with the center of Circle P .

A **tangent** is a line or segment that intersects a circle at exactly one point.

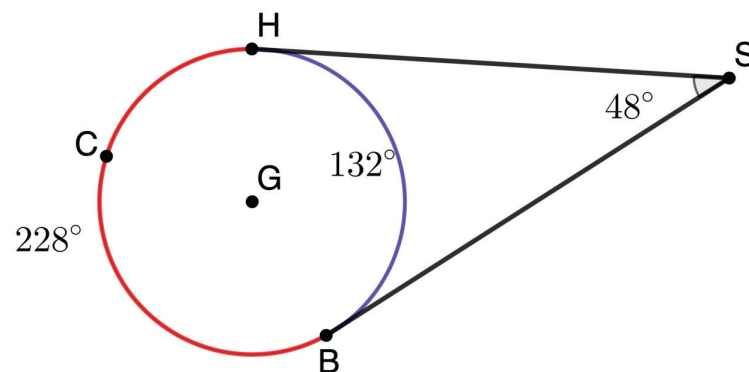


If a line is tangent to a circle, then it is perpendicular to a radius, which ends at the point of tangency.



10.9.3 Collaboration: Angles Formed by Tangents and Secants

1. Circle G with tangents $\overline{SH}$ and $\overline{SB}$ is shown. Use the diagram to answer the following.



- a. Complete the table.

	Measure
Major Arc HCB	
Minor Arc HB	
$\angle HSB$	

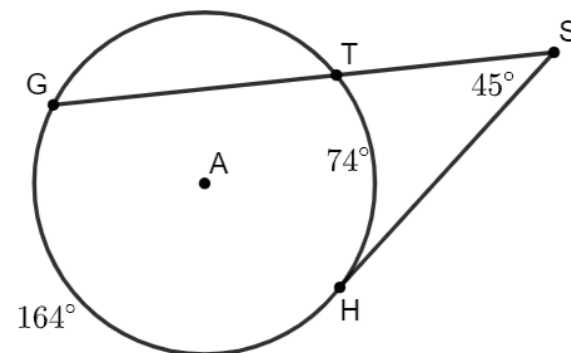
- b. Use the values for the major and minor arcs of Circle G to determine a relationship between the arcs and the measure of the exterior angle formed by the two tangents. The relationship can be described using words or equations.

- c. Ask two other partner pairs to share their relationships. Record their relationships and ask them to initial verifying you have written it correctly. If their description or equation is different than yours, discuss the differences and determine if your description or equation needs to be revised.

Description or Equation	Initials

- d. Explain how to find the measure of an angle formed by two tangents drawn to a circle from a point outside that circle. Be sure to use the term(s) "intercepted arc(s)" at least once in the description.

2. Circle A shows $\angle GSH$ formed by the intersection of tangent $\overline{SH}$ and secant $\overline{GS}$ outside the circle at point S . $m\widehat{GH} = 164^\circ$, $m\widehat{TH} = 74^\circ$, and $m\angle GSH = 45^\circ$.



- a. Use the values for $\widehat{GH}$ and $\widehat{TH}$ of Circle A to determine the relationship between the arcs and the measure of the exterior angle formed by the tangent and the secant.

- b. Discuss the relationship with your partner.



The formula for the measure of an **External Angle to a Circle** formed by two tangents, two secants, or a tangent and a secant is

$$\text{Measure of angle formed} = \frac{1}{2} (\text{difference of intercepted arcs})$$

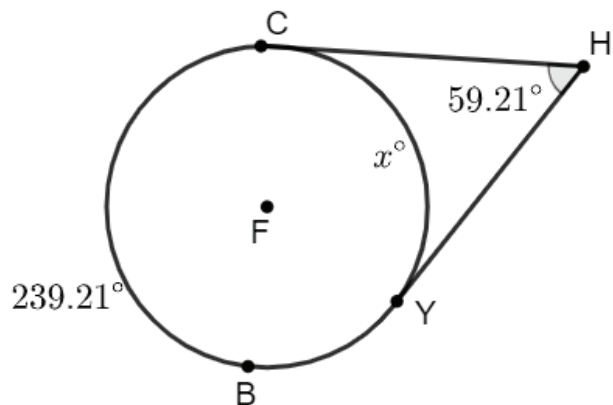
OR

$$\text{Measure of angle formed} = \frac{1}{2} (\text{larger arc} - \text{smaller arc})$$

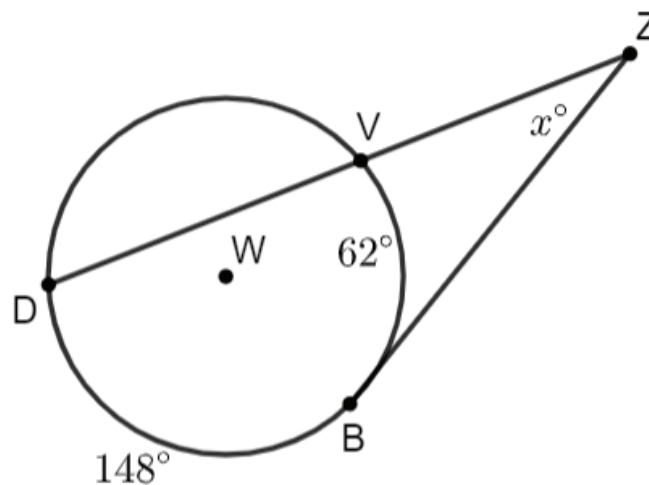


10.9.4 Your Turn!

1. Circle F has points C and Y on the circle. Tangents $\overline{CH}$ and $\overline{YH}$ intersect at point H outside the circle. $m\widehat{CBY} = 239.21^\circ$, $m\widehat{CY} = x^\circ$, and $m\angle CHY = 59.21^\circ$. Find the value of x .



2. Circle W has points D, V , and B on the circle. Tangent $\overline{BZ}$ and secant $\overline{DZ}$ intersect at Point Z outside the circle. $m\widehat{DB} = 148^\circ$, $m\widehat{VB} = 62^\circ$, and $m\angle DZB = x^\circ$. What is $m\angle DZB$?





10.9.5 Wrap-Up: Reflection

Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentences by checking all that apply.

1. I don't understand

- ☐ arc length
- ☐ area of a sector
- ☐ converting between radians and degrees
- ☐ measures of arcs and related angles

...YET.

2. I can explain

- ☐ how to solve problems involving arc length.
- ☐ how to solve problems involving: area of a sector.
- ☐ converting angle measures between degrees and radians.
- ☐ how to solve problems involving measures of arcs and related angles including central, inscribed, and intersections of a chords, secants, and tangents.

Unit 10, Lesson H1: Constructing Tangent Lines



Benchmark

MA.912.GR.5.5 Given a point outside a circle, construct a line tangent to the circle that passes through the given point.



Learning Target

- ☐ I can construct a line tangent to a circle that passes through a given point outside the circle.



10.H1.1 Warm-Up: Radio Signals

- Two-way radio communications help climbers stay in touch with base camp and contact the other expeditioners climbing Mt. Everest. The separate expeditions reserve a central channel where the various groups can share information on weather conditions and route changes or call for rescue.

The equation shows the maximum theoretical range of a radio signal, in miles, using a line-of-sight calculation, which considers the curvature of the Earth and presumes the earth to be a perfect sphere – which of course it is not. This also assumes the radio signal is transmitted with sufficient power that maintains enough energy to travel the distance.

$$\text{Horizon(mi)} = 1.23 \times \sqrt{\text{height(ft)}}$$

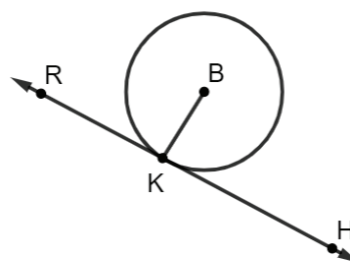


- Use the formula to find the distance the radio signal would go based on your height.
- Explain the relationship of the radio signal range (red line) and Earth.



10.H1.2 Collaboration: Tangent Lines of Circles

- Circle B is shown, with point K on the circle and $\overleftrightarrow{RK}$ tangent to circle B .



- Write two facts about circle B and tangent line $\overleftrightarrow{RK}$.

1.

2.

- Share your facts with your partner and write down one additional fact.

3.

- Complete the statement.

A line tangent to a circle touches a circle at point(s), and the line is _____ to the radius of the circle.

- ☐ one
- ☐ two
- ☐ three

- ☐ parallel
- ☐ perpendicular

to the

3. To construct a line tangent to a circle, you need a compass and a ruler.
- a. Work with your partner and use the steps below to construct a line that is tangent to circle S and goes through point A .



1.	Use the ruler to draw a line from the center, point S , to the given point.
2.	Construct the perpendicular bisector of $\overline{SA}$ to find its midpoint.
3.	Set the compass width to the distance from the midpoint of $\overline{SA}$ to the circle's center, point S .
4.	Keeping the compass at the midpoint, mark an arc across the circle at two places, creating M and V .
5.	Draw the two tangent lines from A to M and A to V .

4. Jamie wrote the steps for constructing a line tangent to a circle on note cards, including the steps of the perpendicular bisector, but the cards got out of order.
- a. Work with your partner to order the steps 1 – 8.

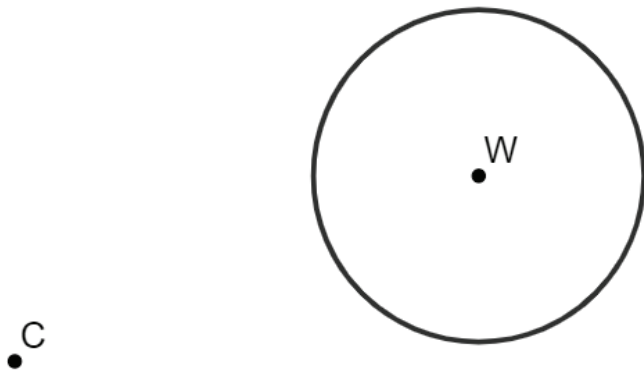
	Place the compass on the midpoint of the line segment and adjust its lengths to reach the center.
	Place the compass at one end of the line segment and adjust the compass to slightly longer than half the line segment.
	Place the ruler where the arcs are above and below the line cross, and draw the perpendicular line.
	Keeping the same width and compass point at the midpoint, draw a circle.
	Draw arcs above and below the line.
	Where the new circle crosses the original circle, create a point and draw the tangent line.
	Draw a line connecting the center of the circle and the point outside the circle.
	Keeping the same compass width, draw arcs above and below the line from the other end of the line segment.

- b. Explain how Jamie's method is different.
- c. Explain why Jamie's method works.



10.H1.3 Your Turn!

1. Given circle W and point C outside the circle, construct a line tangent to circle W going through point using Jamie's method.



2. Naomi and Xavier were practicing constructing a line tangent to circle B going through point A , but their results did not match.

Naomi's Construction	Xavier's Construction

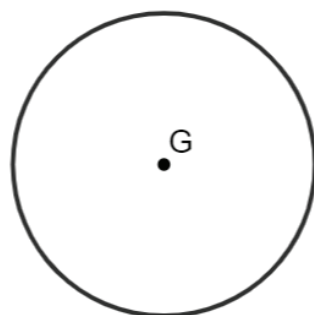
- a. Who correctly drew a line tangent to circle B going through point A ?
- b. Explain the mistake the other student made.



10.H1.4 Wrap-Up: Ticket Out the Door

1. Given circle G and point R outside the circle, construct a line tangent to circle G going through point R .

R



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